EDUGENIE: GOOGLE GEMINI POWERED LEARNING ASSISTANT

Phase 2: Requirement Analysis Phase

The Requirement Analysis Phase focused on identifying the functional and technical requirements needed to implement EduGenie.

Functional Requirements

EduGenie should provide the following functions:

- **Q&A:** Answer academic and general knowledge questions.
- **Explanation:** Explain complex topics in simple language.
- **Quiz:** Generate three multiple-choice questions with four options each.
- **Summary:** Convert lengthy paragraphs into concise summaries.
- **Learning Path:** Provide structured learning recommendations from beginner to advanced levels.

Technologies Required

The following technologies were identified for development:

- Python
- FastAPI
- Uvicorn
- Jinja2
- HTML

- CSS
- Gemini API
- LaMini-Flan-T5-783M

Uvicorn

Uvicorn is used as the ASGI server for running the FastAPI application.

Installation:

```
pip install uvicorn
```

Jinja2

Jinja2 is used as the HTML templating engine for the web interface.

Installation:

```
pip install jinja2
```

AI Model Requirements

Gemini 1.5 Pro is used for:

- Question answering
- Summarization
- Quiz generation
- Learning path generation

LaMini-Flan-T5-783M is used for:

- Concept explanation

The LaMini-Flan-T5 model was selected because it is lightweight, instruction-tuned, and suitable for local execution.